

SE LAB – 1

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Scenario Number – 10

FUNCTIONAL REQUIREMENTS :-

ID	PRIORITY	DESCRIPTION
FR-001	High	Multi-stage budget approval workflow (Faculty Coordinator → Finance Office → Dean)
FR-002	High	Club Lead submits event proposal with attached budget request
FR-003	Medium	System generates a unique, tamper-resistant QR ticket per registered attendee
FR-004	High	System validates scanned tickets at check-in, rejecting duplicates
FR-005	Medium	Campus Admin views master calendar to flag scheduling conflicts

NON-FUNCTIONAL REQUIREMENTS :-

ID	TYPE	PRIORITY	DESCRIPTION
NFR-001	Performance & Security	High	QR scanner validates authenticity in under 100ms
NFR-002	Security & Availability	High	RBAC on approval chain + immutable sign-off logging

UML USE CASE DIAGRAM :-

The use-case diagram models the **Student Club Event Ticketing & Budget Portal** with seven actors and ten use cases.

Actors: Club Lead and Attendee interact from outside the system on the left; Faculty Coordinator, Finance Office, Dean, Campus Admin, and Gate Staff interact from the right. Faculty Coordinator, Finance Office, and Dean are generalized into an abstract **«Approver»** actor, since all three perform the same action (approving a stage of the budget workflow) — this avoids drawing three identical association lines to the same use case.

Use cases and associations:

- Club Lead — Submit Event Proposal, Generate QR Ticket
- Attendee — Register for Event
- «Approver» — Approve Budget Proposal
- Campus Admin — Manage Master Event Calendar

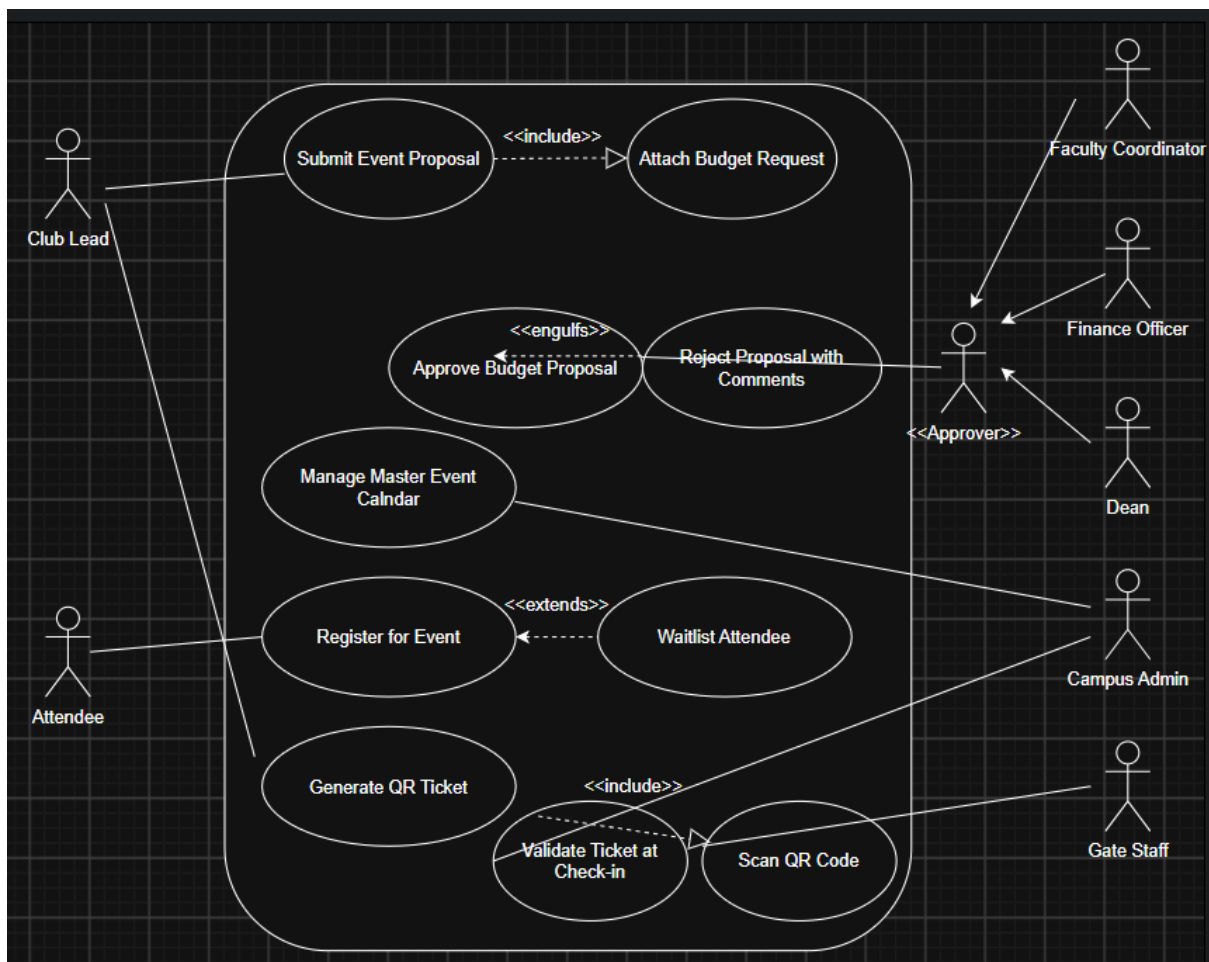
- Gate Staff — Validate Ticket at Check-in

«include» relationships (mandatory sub-steps, always executed):

- Submit Event Proposal includes Attach Budget Request
- Validate Ticket at Check-in includes Scan QR Code

«extend» relationships (optional/conditional sub-flows, only triggered under a condition):

- Reject Proposal with Comments extends Approve Budget Proposal (triggered when an approver rejects instead of signing off)
- Waitlist Attendee extends Register for Event (triggered when the event is full)



USE-CASE FLOW SPECIFICATION :-

Use Case: Approve Budget Proposal

Primary Actor: Approver (generalized: Faculty Coordinator, Finance Office, Dean)

Secondary Actors: Club Lead (notified of outcome), Campus Admin (monitors status)

Related Requirements: FR-001, NFR-002

Brief Description: A submitted event budget proposal is routed sequentially through three approval stages — Faculty Coordinator, then Finance Office, then Dean. Each stage's approver reviews the proposal and either signs off, advancing it to the next stage, or rejects it with comments, which sends it back to the Club Lead (this rejection path is the «extend» use case Reject Proposal with Comments).

Preconditions:

- The event proposal exists in the system with status "Pending Faculty Review" (i.e. Submit Event Proposal, including Attach Budget Request, has completed successfully).
- The logged-in user holds the Approver role matching the proposal's current pending stage (enforced by NFR-002 RBAC).

Postconditions:

- Success: Proposal status is "Approved" and the Club Lead is notified; the system is now authorized to proceed to ticket generation for the event.
- Failure/Alternate: Proposal status is "Rejected – Revision Requested", comments are attached, and the Club Lead is notified; the proposal exits the approval chain until resubmitted.

Main Success Scenario:

1. Approver opens the list of proposals pending their sign-off.
2. Approver selects a proposal and reviews event details and the attached budget request.
3. Approver digitally signs off on the proposal.
4. System validates that the sign-off comes from the actor matching the current stage.
5. System advances the proposal status to the next stage in the hierarchy (Faculty Coordinator → Finance Office → Dean).
6. System logs the sign-off action immutably (actor, timestamp, stage).
7. If the stage just completed was Dean, the system sets status to "Approved" and notifies the Club Lead; use case ends.
8. Otherwise, the system notifies the next-stage Approver that a proposal is now pending their review; use case ends.

Alternate Flow: Rejection at Any Stage (extends Approve Budget Proposal)

- Trigger: At step 3, the Approver chooses to reject instead of sign off.
- A1. Approver enters a mandatory comment explaining the reason for rejection.
- A2. System sets proposal status to "Rejected – Revision Requested", records the comment and the rejecting stage, and logs the action immutably.
- A3. System notifies the Club Lead with the rejection comments.

- A4. Club Lead may revise and resubmit the proposal, which restarts the flow at "Pending Faculty Review" (step 1).
- Use case ends (rejected branch).

Exception Flow: Out-of-Turn Approval Attempt

- Trigger: At step 4, the signing actor does not match the proposal's current pending stage (e.g. Dean attempts to sign off before Finance Office).
- System rejects the sign-off, logs the unauthorized attempt (NFR-002), and displays an error to the actor. Proposal status is unchanged. Use case ends without state change.